```
# Simple Reflex Agent - Vacuum Cleaner Problem

environment = {
    'A': 'Dirty',
    'B': 'Dirty'
}

# Initial position
location = 'A'

# Path cost
path_cost = 0


def reflex_agent(location, status):

    if status == "Dirty":
        return "Suck"

    elif location == "A":
        return "Right"

    else:
        return "Left"


print("Initial Environment:", environment)

while True:
```

```
    status = environment[location]

    action = reflex_agent(location, status)

    print(f"\nVacuum is at {location}")
    print(f"Room Status : {status}")
    print(f"Action Taken: {action}")

    if action == "Suck":

        environment[location] = "Clean"
        path_cost += 1

    elif action == "Right":

        location = "B"
        path_cost += 1

    elif action == "Left":

        location = "A"
        path_cost += 1

    # Check whether both rooms are clean
    if environment["A"] == "Clean" and environment["B"] == "Clean":

        print("\nBoth rooms are clean.")
        break



print("\nFinal Environment:", environment)
```

```
print("Total Path Cost:", path_cost)
```

#RUN

vacuum_cleaner.py